

## Unit-V

### Small Sample tests

#### t-distribution (or) Student's t-distribution:

This distribution is used for testing of hypothesis when the Sample Size is small and population S.D.  $\sigma$  is not known.

Degree of freedom (d.f.) :- It is a number which indicates how many of the values of a variable may be independently (or) freely chosen. For example, if we have to choose any four numbers freely, then we may choose 11, 6, 28, 44 or any other set of four numbers. As in this case all the four numbers have freedom to vary. We say that the degrees of freedom is 4. If we impose a restriction on the numbers, say the sum is 50, then we can choose first three numbers freely and the fourth number is such that the sum is 50. Thus, the three variables are free and independent choices for finding the fourth. Hence these are the degree of freedom. In this case, the degrees of freedom is 3, i.e.,  $4-1$

In general, the number of degrees of freedom is equal to the total number of observations less the number of independent constraints imposed on the observations. For example, in a set of data of  $n$  observations, if  $K$  is the number of independent constraints then  $\nu = n - K$

Definition :- Degrees of freedom is usually denoted by the letter  $\nu$  ('NU' of the Greek alphabet). Suppose we are given a set of  $n$  observations, which are subject to  $K$  independent constraints (restrictions) then  $d.f = \text{No. of observations} - \text{No. of independent constraints on them}$   
i.e.,  $\boxed{\nu = n - K}$

t-distribution :- If  $x_1, x_2, x_3, \dots, x_n$  be any random sample of size  $n$  drawn from a normal population with mean ( $\mu$ ) and variance ( $\sigma^2$ ) then the test statistic 't' is defined

by  $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$  where  $\bar{x}$  = Sample mean and

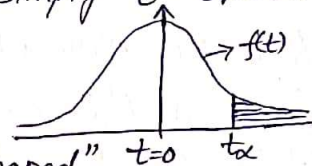
$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$  is an unbiased estimate of  $\sigma^2$

The test statistic  $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$  is a random variable having the  $t$ -distribution with  $\nu = n-1$  degrees of freedom and with probability density function  $f(t) = y_0 \left(1 + \frac{t^2}{\nu}\right)^{\frac{\nu+1}{2}}$

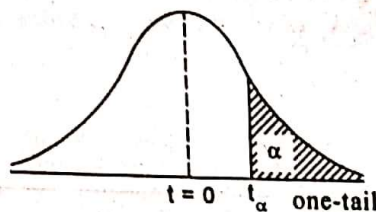
Where  $\nu = n-1$  and  $y_0$  is a constant got by  $\int_{-\infty}^{\infty} f(t) dt = 1$ .

This is known as Student's  $t$ -dist (or) Simply  $t$ -distribution.

Properties of  $t$ -distribution :-



1. The shape of  $t$ -distribution is "bell shaped," which is similar to that of a normal distribution and is symmetrical about the mean.
2. The  $t$ -distribution curve is also asymptotic to the  $t$ -axis, i.e., the two tails of the curve on both sides of  $t=0$  extend to infinity.
3. It is symmetrical about the line  $t=0$ .
4. The form of the probability curve varies with degrees of freedom i.e., with sample size.
5. It is unimodal with Mean = Median = Mode.
6. The mean of standard normal distribution and as well as  $t$ -distribution is zero but variance of  $t$ -distribution depends upon the parameter  $\nu$  which is called the degrees of freedom.
7. The variance of  $t$ -distribution exceeds 1, but approaches 1 as  $n \rightarrow \infty$ . In fact the  $t$ -distribution with  $\nu$ -degrees of freedom approaches standard normal distribution as  $\nu = n-1 \rightarrow \infty$ .

$t_{\alpha}$  - Critical Values of the t-Distribution with  $\nu$  Degrees of Freedom Table - 4

| $\nu$    | $\alpha$ |       |       |       |       |       |        |        |        |
|----------|----------|-------|-------|-------|-------|-------|--------|--------|--------|
|          | 0.40     | 0.30  | 0.20  | 0.15  | 0.10  | 0.05  | 0.025  | 0.01   | 0.005  |
| 1        | 0.325    | 0.727 | 1.376 | 1.963 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 |
| 2        | 0.289    | 0.617 | 1.061 | 1.386 | 1.886 | 2.920 | 4.303  | 6.965  | 9.925  |
| 3        | 0.277    | 0.584 | 0.978 | 1.250 | 1.638 | 2.353 | 3.182  | 4.541  | 5.841  |
| 4        | 0.271    | 0.569 | 0.941 | 1.190 | 1.533 | 2.132 | 2.776  | 3.747  | 4.604  |
| 5        | 0.267    | 0.559 | 0.920 | 1.156 | 1.476 | 2.015 | 2.571  | 3.365  | 4.032  |
| 6        | 0.265    | 0.553 | 0.906 | 1.134 | 1.440 | 1.943 | 2.447  | 3.143  | 3.707  |
| 7        | 0.263    | 0.549 | 0.896 | 1.119 | 1.415 | 1.895 | 2.365  | 2.998  | 3.499  |
| 8        | 0.262    | 0.546 | 0.889 | 1.108 | 1.397 | 1.860 | 2.306  | 2.896  | 3.355  |
| 9        | 0.261    | 0.543 | 0.883 | 1.100 | 1.383 | 1.833 | 2.262  | 2.821  | 3.250  |
| 10       | 0.260    | 0.542 | 0.879 | 1.093 | 1.372 | 1.812 | 2.228  | 2.764  | 3.169  |
| 11       | 0.260    | 0.540 | 0.876 | 1.088 | 1.363 | 1.796 | 2.201  | 2.718  | 3.106  |
| 12       | 0.259    | 0.539 | 0.873 | 1.083 | 1.356 | 1.782 | 2.179  | 2.681  | 3.055  |
| 13       | 0.259    | 0.537 | 0.870 | 1.079 | 1.350 | 1.771 | 2.160  | 2.650  | 3.012  |
| 14       | 0.258    | 0.537 | 0.868 | 1.076 | 1.345 | 1.761 | 2.145  | 2.624  | 2.977  |
| 15       | 0.258    | 0.536 | 0.866 | 1.074 | 1.341 | 1.753 | 2.131  | 2.602  | 2.947  |
| 16       | 0.258    | 0.535 | 0.865 | 1.071 | 1.337 | 1.746 | 2.120  | 2.583  | 2.921  |
| 17       | 0.257    | 0.534 | 0.863 | 1.069 | 1.333 | 1.740 | 2.110  | 2.567  | 2.898  |
| 18       | 0.257    | 0.534 | 0.862 | 1.067 | 1.330 | 1.734 | 2.101  | 2.552  | 2.878  |
| 19       | 0.257    | 0.533 | 0.861 | 1.066 | 1.328 | 1.729 | 2.093  | 2.539  | 2.861  |
| 20       | 0.257    | 0.533 | 0.860 | 1.064 | 1.325 | 1.725 | 2.086  | 2.528  | 2.845  |
| 21       | 0.257    | 0.532 | 0.859 | 1.063 | 1.323 | 1.721 | 2.080  | 2.518  | 2.831  |
| 22       | 0.256    | 0.532 | 0.858 | 1.061 | 1.321 | 1.717 | 2.074  | 2.508  | 2.819  |
| 23       | 0.256    | 0.532 | 0.858 | 1.060 | 1.319 | 1.714 | 2.069  | 2.500  | 2.807  |
| 24       | 0.256    | 0.531 | 0.857 | 1.059 | 1.318 | 1.711 | 2.064  | 2.492  | 2.797  |
| 25       | 0.256    | 0.531 | 0.856 | 1.058 | 1.316 | 1.708 | 2.060  | 2.485  | 2.787  |
| 26       | 0.256    | 0.531 | 0.856 | 1.058 | 1.315 | 1.706 | 2.056  | 2.479  | 2.779  |
| 27       | 0.256    | 0.531 | 0.855 | 1.057 | 1.314 | 1.703 | 2.052  | 2.473  | 2.771  |
| 28       | 0.256    | 0.530 | 0.855 | 1.056 | 1.313 | 1.701 | 2.048  | 2.467  | 2.763  |
| 29       | 0.256    | 0.530 | 0.854 | 1.055 | 1.311 | 1.699 | 2.045  | 2.462  | 2.756  |
| 30       | 0.256    | 0.530 | 0.854 | 1.055 | 1.310 | 1.697 | 2.042  | 2.457  | 2.750  |
| 40       | 0.255    | 0.529 | 0.851 | 1.050 | 1.303 | 1.684 | 2.021  | 2.423  | 2.704  |
| 60       | 0.254    | 0.527 | 0.848 | 1.045 | 1.296 | 1.671 | 2.000  | 2.390  | 2.660  |
| 120      | 0.254    | 0.526 | 0.845 | 1.041 | 1.289 | 1.658 | 1.980  | 2.358  | 2.617  |
| $\infty$ | 0.253    | 0.524 | 0.842 | 1.036 | 1.282 | 1.645 | 1.960  | 2.326  | 2.576  |

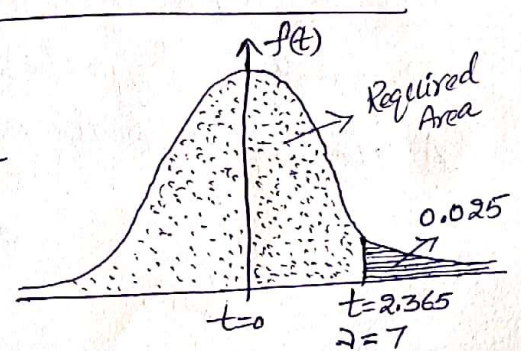
**Note :** The above table gives the values of  $t$  for one-tail test (either left-tail or right-tail test). If we have to find the value of  $t$  for a two-tail test at a level, we take the value of  $\alpha / 2$  for  $\alpha$ . For example, the value of  $t$  at 5% level with 9 d.f. is  $t_{0.025} = 2.262$  and the value of  $t$  at 1% level with 11 d.f. is  $t_{0.005} = 3.106$

## Applications of the t-distribution :-

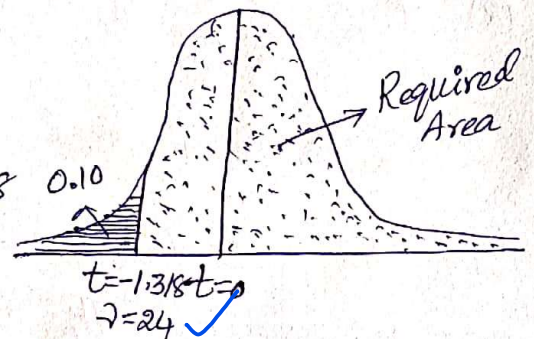
The t-distribution has a wide number of applications in statistics, Some of them are given below.

1. To test the Significance of the Sample mean, When population variance is not given.
2. To test the Significance of the mean of the Sample i.e., to test if the Sample mean differs Significantly from the population mean.
3. To test the Significance of the difference between two Sample means or to Compare two Samples.
4. To test the Significance of an Observed Sample Correlation Coefficient and Sample regression Coefficient.

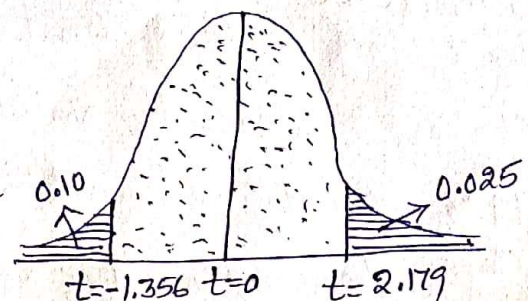
(a) When  $t < 2.365$ ,  $P(t < 2.365)$  is given by the area to the left of  $t = 2.365$   
From table  $t_{\alpha} = 2.365$  for  $\gamma = 7$  d.f  
then  $\alpha = 0.025$   
 $\therefore P(t < 2.365) = 1 - 0.025 = 0.975$



(b) When  $t > -1.318$   
 $P(t > -1.318)$  is given by the area to the right of  $t = -1.318$  From table  $t_{\alpha} = 1.318$   
With  $\gamma = 24$  d.f. then  $\alpha = 0.10$



(c) When  $t < 2.179$  with  $\gamma = 12$  d.f  
Area to the right of 2.179 is 0.025  
When  $t > -1.356$  with  $\gamma = 12$  d.f  
Area to the left of 1.356 is 0.10  
 $\therefore$  When  $-1.356 < t < 2.179$ , the area is  $1 - 0.10 - 0.025 = 0.875$   
 $\therefore P(-1.356 < t < 2.179) = 0.875$



## Student's 't' test for Single mean

Suppose we want to test

- (a) If a random sample  $x_i$  of size  $n$  has been drawn from a normal population with a specified mean  $\mu$ .
- (b) If the sample mean differs significantly from the hypothetical value  $\mu$  of the population mean

In this case the statistic is given by  $t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{(n-1)} \text{ d.f.}$

Where  $\bar{x}$ ,  $\mu$ ,  $s$ ,  $n$  have usual meanings.

Let a random sample of size  $n$  ( $n < 30$ ) has a sample mean  $\bar{x}$ . To test the hypothesis that the population mean  $\mu$  has a specified value  $\mu_0$  when population S.D ( $\sigma$ ) is not known.

Let the null hypothesis be  $H_0: \mu = \mu_0$  then

Alternative hypothesis is  $H_1: \mu \neq \mu_0$

Assuming that  $H_0$  is true, the test statistic given by

(1)  $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$ ,  $S$  - Sample S.D is given directly.

(2)  $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$ ,  $S$  - Sample S.D is ~~not~~ not given directly, which can be calculated by the given samples.

With  $(n-1)$  d.f we can calculate the value of  $t_{\alpha}$  at  $\alpha$

If the calculated value of  $t >$  the table value of  $t$

We reject  $H_0$  at  $\alpha$  otherwise we Accept  $H_0$ .

**Confidence limits for population mean ( $\mu$ ):-**

(i) 95% Confidence limits for  $\mu$  is  $\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}}$

ie  $t_{\alpha/2}$  value depends upon Sample Size  $n$  and  $\alpha = n-1$

(ii) 99% Confidence limits for  $\mu$  is  $\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}}$

here also  $t_{\alpha/2}$  value depends upon the Sample Size  $n$ , and  $\alpha = n-1$  d.f

Problem:- (1):- A Sample of 26 bulbs gives a mean life of 990 hrs with a S.D of 20 hrs. The manufacturer claims that the mean life of bulbs is 1000 hours. Is the Sample not upto the standard?

Solution:- Given Sample Size  $n = 26 < 30$  (Small Sample)

Sample mean  $\bar{x} = 990$  hours

Sample S.D  $S = 20$  hours.

population mean  $\mu = 1000$  hours.

We can use student's t-test

(1) Null hypothesis  $H_0$ : Sample is upto the standard ( $\mu = 1000$ )

(2) Alternative hypothesis  $H_1$ : Sample is below the standard ( $\mu < 1000$ )  
(left tailed test.)

(3) Level of Significance:  $\alpha = 0.05$  (assume)

degrees of freedom:  $\gamma = n - 1 \Rightarrow \gamma = 26 - 1 = 25$

$t_\alpha$  value at  $\alpha = 0.05$  and 25 d.f is 1.708 i.e.,  $t_\alpha = 1.708$

(4) The test statistic:  $t = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n-1}}} = \frac{990 - 1000}{\frac{20}{\sqrt{26-1}}} = -2.5$

$$\therefore |t| = 2.5$$

(5) Conclusion:-  $|t| = 2.5$ ,  $t_\alpha = 1.708$

$\therefore |t| > t_\alpha$  : We Reject  $H_0$

We Conclude that the Sample is not upto the standard.

Problem@:- A random Sample of Six steel beams has a mean Compressive strength of 58392 P.s.i (pounds per Square inch) with a Standard deviation of 648 P.s.i. Use this information and the level of significance  $\alpha = 0.05$  to test whether the true average Compressive strength of the steel from which this Sample came is 58000 P.s.i. Assume normality.

Solution:- We have Sample Size  $n = 6$

Sample mean  $\bar{x} = 58392$  P.s.i

Sample S.D  $S = 648$  P.s.i

Population mean  $\mu = 58000$  P.S.i

In this problem  $\sigma$  is not known and  $n < 30$ . Hence we use t-test

- 1). Null hypothesis  $H_0: \mu = 58000$
- 2) Alternative hypothesis  $H_1: \mu \neq 58000$  (two tailed test)
- 3). L.O.S:  $\alpha = 0.05$  ; degrees of freedom  $\gamma = n-1 \Rightarrow \gamma = 6-1 = 5$

$t_{\alpha/2}$  value at  $\alpha = 0.05$  with 5 d.f is 2.571

ie  $t_{\alpha/2} = 2.571$

(4) The test statistic:  $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{58392 - 58000}{648/\sqrt{6-1}} = 1.353$

$|t| = 1.353$

(5) Conclusion:-  $|t| = 1.353$ ,  $t_{\alpha/2} = 2.571$

$|t| < t_{\alpha} \therefore$  we Accept  $H_0$

Hence the average Compressive Strength of the Steel beam is equal to 58000 P.S.i

Problem (3):- A Random Sample of size 16 values from a normal population showed a mean of 53 and a Sum of Squares of deviations from the mean equals to 150. Can this Sample be regarded as taken from the population having 56 as mean? obtain 95% Confidence limits of the mean of the population.

Solution:- Given Sample Size  $n = 16$   
Sample mean  $\bar{x} = 53$

Sum of squares of deviations from mean is  $\sum (x_i - \bar{x})^2 = 150$

$\therefore s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{150}{15} = 10$        $s = \sqrt{10}$

- 1) Null hypothesis  $H_0$ : The Sample is taken from the population having 56 as mean ie  $\mu = 56$
- 2) Alternative hypothesis:  $H_1: \mu \neq 56$  (two tailed test)

3). Level of Significance:  $\alpha = 0.05$

$t_{\alpha/2}$  value at  $\alpha=0.05$  with 15 d.f is given by 2.13

$$t_{\alpha} = 2.13$$

4). The test Statistic:  $t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{53 - 56}{\sqrt{10}/\sqrt{16}} = -3.79$

$$\therefore |t| = 3.79$$

5). Conclusion:-  $|t| = 3.79$  ;  $t_{\alpha} = 2.13$

$|t| > t_{\alpha}$   $\therefore$  we Reject the null hypothesis  $H_0$

$\therefore$  we Conclude that the Sample Cannot be regarded as taken from the population.

95% Confidence limits for the population mean are given by

$$\bar{x} \pm t_{0.05} \frac{s}{\sqrt{n}} \\ = 53 \pm 2.131 \times \frac{\sqrt{10}}{\sqrt{16}} = 53 \pm (2.13)(0.79) = 53 \pm 1.6827$$

$$\text{ie } 53 - 1.6827 \leq \mu \leq 53 + 1.6827$$

$$51.3173 \leq \mu \leq 54.6827$$

Hence 95% Confidence limits are  $[51.32, 54.68]$

Problem (4):- A random Sample of 10 boys had the following

I.Q's : 70, 120, 110, 101, 88, 83, 95, 98, 107 and 100. (a) Do these data Support the assumption of a population mean I.Q's of 100 ?  
(b) Find a reasonable range in which most of the mean I.Q values of Samples of 10 boys lie.

Solution:- (a) Here S.D and mean of the sample is not given directly. we have to determine these S.D and mean as follows.

$$\text{Mean } \bar{x} = \frac{\sum x_i}{n} = \frac{70+120+\dots+100}{10} = \frac{972}{10} = 97.2$$

$$\text{We know that } s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{1833.60}{10-1} = \frac{1833.6}{9} = 203.73$$

$$s = \sqrt{203.73} = 14.27$$

$$S = 14.27$$

1). Null hypothesis  $H_0$ : The data support the assumption of a population mean I.Q of 100 in the population.

$$\text{ie } H_0: \mu = 100$$

2). Alternative hypothesis  $H_1$ :  $\mu \neq 100$  (two tailed test)

3). Level of Significance:  $\alpha = 0.05$  (ie  $\alpha/2 = \frac{0.05}{2} = 0.025 = \alpha$ )

Critical value ~~ta~~ at  $\alpha = 0.05$  with 9 d.f is 2.26

$$\text{ie, } t_{\alpha} = 2.26$$

4). Test statistic:  $t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{97.2 - 100}{14.27/\sqrt{10}} = -0.62$

$$|t| = 0.62 \text{ ie calculated value of } t = 0.62$$

5) Conclusion:-  $|t| = 0.62$  ;  $t_{\alpha} = 2.26$

$$|t| < t_{\alpha} \therefore \text{We Accept } H_0.$$

We conclude that the data support the assumption of mean I.Q of 100 in the population.

(b) The 95% Confidence limits are given by  $\bar{x} \pm t_{0.05} \frac{s}{\sqrt{n}}$

$$97.5 - (2.26) \left( \frac{14.27}{\sqrt{10}} \right) \leq \mu \leq 97.5 + (2.26) \left( \frac{14.27}{\sqrt{10}} \right)$$

$$97.5 - (2.26)(4.513) \leq \mu \leq 97.5 + (2.26)(4.513)$$

$$97.5 - 10.198 \leq \mu \leq 97.5 + 10.198$$

$$87.302 \leq \mu \leq 107.69$$

$\therefore$  The 95% Confidence limits within which the mean I.Q values of samples of 10 boys will lie is (87.302, 107.69)

Problem 5:- The life time of electric bulbs for a random sample of 10 from a large Consignment gave the following data.

|                  |   |     |     |     |     |     |     |     |     |     |     |
|------------------|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Item             | : | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  |
| Life in 1000 hrs | : | 1.2 | 4.6 | 3.9 | 4.1 | 5.2 | 3.8 | 3.9 | 4.3 | 4.4 | 5.6 |

Can we accept the hypothesis that the average life time of bulbs is 4000 hrs.

Solution:- Given, Sample Size  $n = 10$

- 1) Null hypothesis  $H_0: \mu = 4000$  hrs
- 2) Alternative hypothesis  $H_1: \mu \neq 4000$  hrs (two tailed test)
- 3) Level of Significance  $\alpha = 0.05$

Critical values  $t_{\alpha}$  for  $\alpha = 0.05$  for two tailed test 0.025 with '9' d.f

$$t_{\alpha/2} = 2.262 \text{ (table value)}$$

$$4) \text{ Test statistic } t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$\bar{x} = \text{average life time of bulbs} = \frac{1.2 + 4.6 + \dots + 5.6}{10} = \frac{41}{10} = 4.1 \approx 4100 \text{ hrs}$$

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{(1.2-4.1)^2 + (4.6-4.1)^2 + \dots + (5.6-4.1)^2}{10-1} = \frac{12.42}{9} = 1.38$$

$$s^2 = 1.38 \approx 1380$$

$$s = \sqrt{1380} = 37.15$$

$$\text{Hence } |t| = \frac{4100 - 4000}{37.15/\sqrt{10}} = \frac{316.2278}{37.15} = 8.512$$

$$5) \text{ Conclusion:- } |t| = 8.512, t_{\alpha} = 2.262$$

$|t| > t_{\alpha} \therefore$  We Reject the null hypothesis  $H_0$

We conclude that the average life time of bulbs is not 4000 hours.

Problem 6:- A random Sample from a Company's very extensive files shows that the orders for a certain kind of machinery were filled, respectively in 10, 12, 19, 14, 15, 18, 11 and 13 days. Use the level of Significance  $\alpha = 0.01$  to test the claim that on the average such orders are filled in 10.5 days. Choose the alternative hypothesis that the rejection of null hypothesis  $\mu = 10.5$  days implies that it takes longer than indicated.

Problem (7):— Producer of 'gutkha', Claims that the nicotine Content in his 'gutkha' on the average is 1.83 mg. Can this claim accepted if a random Sample of 8 'gutkha' of this type have the nicotine Contents of 2.0, 1.7, 2.1, 1.9, 2.2, 2.1, 2.0, 1.6 mg. Use a 0.05 level of Significance.

Problem (8):— A random Sample of size 25 from a normal population has the mean  $\bar{x} = 47.5$  and the standard deviation  $S = 8.4$ . Does this information tend to Support (or) refuse the claim that the mean of the population is  $\mu = 42.5$ ?

Problem (9):— The mean life time of a Sample of 25 fluorescent light bulbs produced by a Company is Computed to be 157 hours with S.D of 120 hours. The Company Claims that the average life of the bulbs produced by the Company is 1600 hours Using the level of Significance of 0.05. Is the Claim acceptable?

Problem (10):— Prices of Shares of a Company on the different days in a month were found to be 66, 65, 69, 70, 69, 71, 70, 63, 63, 64, and 68. Discuss whether the mean price of the shares in the month is 65.

### Student's t - test for Difference of means

Let  $\bar{x}$  and  $\bar{y}$  be the means of two independent Samples of sizes  $n_1$  and  $n_2$  ( $n_1 < 30, n_2 < 30$ ) drawn from two normal populations having means  $\mu_1$  and  $\mu_2$ . To test whether the two populations means are equal (ie., to test whether the difference  $\mu_1 - \mu_2$  is Significant.)

Let the Null hypothesis  $H_0: \mu_1 = \mu_2$

Then Alternative hypothesis  $H_1: \mu_1 \neq \mu_2$

If  $\sigma_1 = \sigma_2 = \sigma$ , then an unbiased estimate  $S^2$  of the Common Variance  $\sigma^2$  is given by  $S^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2}$ . Where  $S_1^2, S_2^2$  are Sample Variances.

also S.E of  $(\bar{x} - \bar{y}) = S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ , where  $S = \sqrt{\frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2}}$

Assuming that  $H_0$  is true, the test statistic  $t$  given by

$$t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{(n_1+n_2-2)} \text{ d.f.}, S^2 = \frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2}$$

95% Confidence limits for the difference of two population means are  $(\bar{x} - \bar{y}) \pm t_{\alpha} \cdot S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ , where  $\alpha = 0.025$   
 and 99% Confidence limits for the difference of two population means are  $(\bar{x} - \bar{y}) \pm t_{\alpha} \cdot S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ , where  $\alpha = 0.005$

Problem ①:- Two horses A and B were tested according to the time (seconds) to run a particular track with the following results.

|          |    |    |    |    |    |    |    |
|----------|----|----|----|----|----|----|----|
| Horse A: | 28 | 30 | 32 | 33 | 33 | 29 | 34 |
| Horse B: | 29 | 30 | 30 | 24 | 27 | 29 | -  |

Test whether the two horses have the same running capacity.

Solution:- Given  $n_1 = 7$ ,  $n_2 = 6$  } We first compute the sample means and standard deviations.

$$\bar{x} = \frac{28+30+\dots+34}{7} = \frac{219}{7} = 31.286$$

$$\bar{y} = \frac{29+30+\dots+29}{6} = \frac{169}{6} = 28.16$$

$$\sum (x_i - \bar{x})^2 = (28 - 31.29)^2 + (30 - 31.29)^2 + \dots + (34 - 31.29)^2 = 31.4358$$

$$\sum (y_i - \bar{y})^2 = (29 - 28.16)^2 + (30 - 28.16)^2 + \dots + (29 - 28.16)^2 = 26.8336$$

$$S^2 = \frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2} = \frac{[31.4358 + 26.8336]}{7+6-2} = \frac{58.2694}{11}$$

$$S^2 = 5.23 \Rightarrow S = \sqrt{5.23} = 2.3$$

Test Procedure:- (1) Null hypothesis  $H_0: \mu_1 = \mu_2$   
 (2) Alternative hypothesis  $H_1: \mu_1 \neq \mu_2$

3) Level of Significance:  $\alpha = 0.05$

Critical value  $t_{\alpha}$  of  $t$  for  $7+6-2=11$  d.o.f at 5% level of Significance is 2.201. ie.,  $t_{\alpha} = 2.201$

4) Test Statistic:  $t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{31.286 - 28.16}{(2.3) \sqrt{\left(\frac{1}{7} + \frac{1}{6}\right)}} = 2.443$

$$|t| = 2.443$$

5) Conclusion:-  $|t| = 2.443$  ;  $t_{\alpha} = 2.201$

$|t| > t_{\alpha} \therefore$  We Reject  $H_0$

$\therefore$  We Conclude that ~~Both~~ horses A and B do not have the same running capacity.

Problem 2:- To examine the hypothesis that the husbands are more intelligent than the wives, an investigator took a sample of 10 couples and administered them a test which measures the I.Q. The results are as follows:

Husbands : 117 105 97 105 123 109 86 78 103 107

Wives : 106 98 87 104 116 95 90 69 108 85

Test the hypothesis with a reasonable test at the level of significance of 0.05.

Solution:- We have  $n_1 = 10$  ,  $n_2 = 10$

$$\bar{x} = \frac{117 + 105 + \dots + 107}{10} = \frac{1030}{10} = 103$$

$$\bar{y} = \frac{106 + 98 + \dots + 85}{10} = \frac{958}{10} = 95.8$$

$$\sum (x_i - \bar{x})^2 = (117 - 103)^2 + (105 - 103)^2 + \dots + (107 - 103)^2 = 1506$$

$$\sum (y_i - \bar{y})^2 = (106 - 95.8)^2 + (98 - 95.8)^2 + \dots + (85 - 95.8)^2 = 1679.6$$

$$\text{Now } S^2 = \frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2} = \frac{1606 + 1679.6}{10 + 10 - 2} = \frac{3285.6}{18} = 182.53$$

$$\therefore S = \sqrt{182.53} = 13.51$$

Test Procedure: 1) Null hypothesis  $H_0: \mu_1 = \mu_2$  (ie, No difference in IQ.)

(2) Alternative hypothesis  $H_1: \mu_1 > \mu_2$  (Right tailed test)  
(ie, husbands are more intelligent than wives)

3). Level of Significance:  $\alpha = 0.05$

Critical Value  $t_\alpha$  of  $\alpha = 0.05$  with 18 d.f is 1.734

$$t_\alpha = 1.734$$

4). Test Statistic:  $t = \frac{\bar{x} - \bar{y}}{S \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{103 - 95.8}{(13.51) \sqrt{\left(\frac{1}{10} + \frac{1}{10}\right)}} = 1.1917$

$$|t| = 1.1917$$

5). Conclusion:-  $|t| = 1.1917$ ;  $t_\alpha = 1.734$

$|t| < t_\alpha \therefore$  we Accept  $H_0$

$\therefore$  We Conclude that husbands and Wives are equally intelligent.

Practice Problems:- (1) The means of two random samples of sizes 9 and 7 are 196.42 and 198.82 respectively. The Sum of Squares of the deviations from the mean are 26.94 and 18.73 respectively. Can the Sample be considered to have been drawn from the Same normal populations.

(2) Measuring specimens of nylon yarn, taken from two machines, it was found that 8 specimens from first machine had a mean denier of 9.67 with a S.D of 1.81 While 10 specimens from Second machines had a mean denier of 7.43 With a S.D of 1.48. Assuming that the proportions are normal, test the hypothesis  $H_0: \mu_1 - \mu_2 = 1.5$ , against alternative hypothesis  $H_1: \mu_1 - \mu_2 > 1.5$  at 0.05 level of significance.

## Paired Sample t-test

If  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  be the pairs of Sales data before and after the sales probation in a business concern, we apply paired t-test to examine the Significance of the difference of the two situations. Let  $d_i = x_i - y_i$  (or)  $y_i - x_i$  for  $i = 1, 2, \dots, n$

Let the Null hypothesis be  $H_0: \mu_1 = \mu_2$  i.e., There is no Significant difference between the means in two situations.

Then Alternative hypothesis  $H_1: \mu_1 \neq \mu_2$ .

Assuming that  $H_0$  is true, the test statistic for  $n$  paired observations (which are dependent) by taking the differences  $d_1, d_2, \dots, d_n$  of the paired data.

$$t = \frac{\bar{d} - \mu}{s/\sqrt{n}} = \frac{\bar{d}}{s/\sqrt{n}} \quad (\because \mu = 0) \quad \bar{d} = \frac{\sum d_i}{n}$$

and  $s^2 = \frac{1}{n-1} \sum (d_i - \bar{d})^2$

$\bar{d}$  and  $s^2$  are the mean and variance of the differences  $d_1, d_2, \dots, d_n$  respectively and  $\mu$  is the mean of the population of differences. and the test statistic follows student's  $t$  distribution with  $(n-1)$  degrees of freedom.

Problem ①:- Scores obtained in a shooting competition by 10 soldiers before and after intensive training are given below.

Before : 67 24 57 55 63 54 56 68 33 43

After : 70 38 58 58 56 67 68 75 42 38

Test whether the intensive training is useful at 0.05 level of significance.

Solution:- let us apply paired test

Let  $\mu$  be the mean of population of differences.

- 1) Null hypothesis  $H_0: \mu_1 = \mu_2$  i.e.  $\mu = 0$  There is no Significant effect of the training
- 2) Alternative hypothesis  $H_1: \mu_1 < \mu_2$  (ie intensive training is useful)
- 3) Level of Significance  $\alpha = 0.05$

$t_{\alpha}$  at  $\alpha = 0.05$  with  $10-1 = 9$  d.o.f is 1.83 i.e.  $t_{\alpha} = 1.83$

4) Test statistic :  $t = \frac{\bar{d}}{s/\sqrt{n}}$

Compute the differences : -3, -14, -1, -3, +7, -13, -12, -7, -9 and 5

$$\bar{d} = \frac{-3-14-1-3+7-13-12-7-9+5}{10} = \frac{-50}{10} = -5$$

$$s^2 = \frac{\sum (d_i - \bar{d})^2}{n-1} = \frac{(-3+5)^2 + (-14+5)^2 + \dots + (5+5)^2}{10-1} = \frac{482}{9} = 53.55$$

$$\therefore s = \sqrt{53.55} = 7.32$$

$$t = \frac{-5-0}{7.32/\sqrt{10}} = -2.16$$

$$\boxed{|t| = 2.16}$$

5) Conclusion:-  $|t| = 2.16$  ;  $t_{\alpha} = 1.83$

$|t| > t_{\alpha} \therefore$  We Reject the Null hypothesis  $H_0$

$\therefore$  We conclude that the intensive training is Useful.

Problem 2:- The Blood Pressure of 5 women before and after intake of a certain drug are given below.

Before: 110 120 123 132 125

After: 120 118 125 136 121

Test whether there is Significant change in Blood Pressure at 1% level of Significance.

Solution:- Let the Null hypothesis  $H_0: \mu_1 = \mu_2$

ie There is no difference Significant in blood pressure before and after intake of drug.

then alternative hypothesis  $H_1: \mu_1 < \mu_2$

Assuming that  $H_0$  is true then the test statistic is

$$t = \frac{\bar{d}}{s/\sqrt{n}}$$

The differences are :-10, +2, -2, -4, 4

$$\bar{d} = \frac{-10 + 2 - 2 + 4 + 4}{5} = -2$$

$$s^2 = \frac{\sum (d_i - \bar{d})^2}{n-1} = \frac{(10+2)^2 + (2+2)^2 + \dots + (4+2)^2}{5-1} = \frac{120}{4} = 30$$

$$s = \sqrt{30} =$$

$$-t = \frac{\bar{d}}{s/\sqrt{n}} = \frac{-2}{\sqrt{30}/\sqrt{5}} = \frac{-2}{\sqrt{6}} = -0.82$$

$$|t| = 0.82$$

Conclusion:-  $|t| = 0.82$ ,  $t_{\alpha} = 4.6$

$|t| < t_{\alpha} \therefore$  We Accept the Null hypothesis  $H_0$ .  
We conclude that there is no significant difference between Blood pressure of before and after intake of a certain drug.

Practice problems:- (1) Memory Capacity of 10 Students were tested before and after training. state whether the training was effective (or) not from the following scores.

Before training: 12 14 11 8 7 10 3 0 5 6

After training: 15 16 10 7 5 12 10 2 3 8

(2) An I.Q test was administered to 5 persons before and after they were trained. the results are given below.

I.Q. Before training: 108 121 123 132 122

I.Q. After training: 118 120 128 137 131

Test whether there is any change in I.Q after the training Programme.

## F-distribution

⑨

(Sampling distribution of the ratio of two Sample Variances)

Another important Continuous probability distribution which plays an important role in connection with Sampling from normal populations is the F-distribution.

Let  $S_1^2$  be the Sample Variance of an independent Sample of Size  $n_1$  drawn from a normal population  $N(\mu_1, \sigma_1^2)$ . Similarly, let  $S_2^2$  be the Sample Variance in an independent Sample of Size  $n_2$  drawn from another normal population  $N(\mu_2, \sigma_2^2)$ .

Thus  $S_1^2$  and  $S_2^2$  are two Variances of two random Samples of Sizes  $n_1$  and  $n_2$  respectively drawn from the normal population, with the variances  $\sigma_1^2$  and  $\sigma_2^2$ . To determine whether the two Samples come from two populations having equal variances.

Consider the Sampling distribution of the ratio of the Variances of the two independent random Samples defined by  $F = \frac{S_1^2}{S_2^2}$ ,  $S_1^2 > S_2^2$

which follows F-distribution with  $r_1 = n_1 - 1$  and  $r_2 = n_2 - 1$  degrees of freedom.

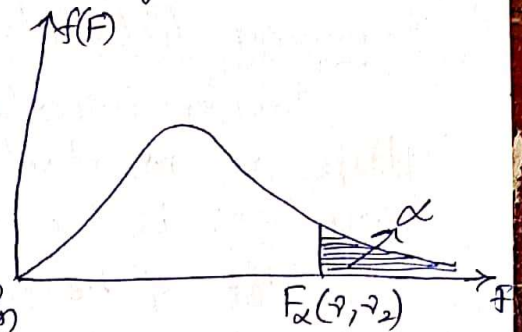
F-distribution can be used to test the equality of several populations means, comparing Sample Variances and analysis of Variance.

Under the assumption (hypothesis) that two normal populations have the same Variance i.e.,  $\sigma_1^2 = \sigma_2^2$

We have  $F = \frac{S_1^2}{S_2^2}$ , which determines whether the ratio of two Sample Variances  $S_1$  and  $S_2$  is too small (or) too large.

$F$  is closer to 1. the two Sample Variances  $S_1$  and  $S_2$  are almost same. In practice, it is customary, to take the larger variance as the numerator.

$F$  is always a positive number.



### Properties of F-distribution:-

- 1). F-distribution is free from population Parameters and depends <sup>upon</sup> degrees of freedom only.
- 2). F-distribution Curve lies entirely in first-quadrant.
- 3). The F-Curve depends not only on the two parameters  $r_1$  and  $r_2$  but also on the order in which they are stated.
- 4).  $F_{(1-\alpha)}(r_1, r_2) = \frac{1}{F_{\alpha}(r_2, r_1)}$  Where  $F_{\alpha}(r_1, r_2)$  is the value of  $F$  with  $r_1$  and  $r_2$  degrees of freedom such that the area under the F-distribution Curve to the right of  $F_{\alpha}$  is  $\alpha$ .
- 5). The mode of F-distribution is less than unity.

### Test for equality of two population Variances

Let two independent random samples of sizes  $n_1$  and  $n_2$  be drawn from two normal populations.

To test the hypothesis that the two population Variances  $\sigma_1^2$  and  $\sigma_2^2$  are equal.

Let the null hypothesis  $H_0: \sigma_1^2 = \sigma_2^2$

Then alternative hypothesis  $H_1: \sigma_1^2 \neq \sigma_2^2$

The estimates of  $\sigma_1^2$  and  $\sigma_2^2$  are given by

$$S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1} \text{ and } S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{\sum (y_i - \bar{y})^2}{n_2 - 1}$$

Where  $s_1^2$  and  $s_2^2$  are the variances of the two samples.

Assuming that  $H_0$  is true, the test statistic

(10)

$$F = \frac{S_1^2}{S_2^2} \text{ (or) } \frac{S_2^2}{S_1^2} \text{ according as } S_1^2 > S_2^2 \text{ (or) } S_2^2 > S_1^2$$

follows F-distribution with  $(n_1-1, n_2-1)$  degrees of freedom.

If the calculated value of  $F >$  the tabulated value of  $F$  at 5% level, we reject the null hypothesis  $H_0$ . Otherwise we ~~reject~~ Accept  $H_0$ .

Note :- ①  $F = \frac{\text{Greater Variance}}{\text{Smaller Variance}}$

(2). When  $F$  is close to 1, the two Sample Variances  $S_1$  and  $S_2$  are nearly same.

3). If Sample Variance  $S^2$  is given, we can obtain population Variance  $\sigma^2$  by using the relation  $n\sigma^2 = (n-1)S^2$  and vice-versa.

4).  $F_\alpha(r_1, r_2)$  is the value of  $F$  with  $r_1$  and  $r_2$  degrees of freedom such that the area under the F-distribution to the right of  $F_\alpha$  is  $\alpha$ . In tables  $F_\alpha$  tabulated for  $\alpha = 0.05$  and  $\alpha = 0.01$  for various combinations of the degrees of freedom  $r_1$  and  $r_2$  clearly value of  $F$  at 5% significance is lower than at 1%.

Problem ① :- If two independent random samples of sizes  $n_1 = 13$  and  $n_2 = 7$  are taken from a normal population. What is the probability that the variance of the first sample will be at least four times as large as that of the second sample?

Solution :- Given,  $n_1 = 13$ ,  $n_2 = 7$

and  $S_1^2 = 4S_2^2$  i.e., the variance of the first sample will be at least four times as large as that of the second sample.

$$\text{Now, } F = \frac{S_1^2}{S_2^2} = \frac{4S_2^2}{S_2^2} = 4.00$$

$$\therefore F = 4.00 \quad (\because S_1^2 = 4S_2^2)$$

This value of  $F$  follows  $F$ -distribution with  $\gamma_1 = n_1 - 1$  and  $\gamma_2 = n_2 - 1$  i.e.  $\gamma_1 = 12$  and  $\gamma_2 = 6$  degrees of freedom. Hence from the table we get  $F_{0.05}(12, 6) = 4.00$ .

Therefore, the required probability is 0.05.

Problem (2):- In one sample of 8 observations from a normal population, the sum of the squares of deviations of the sample values from the sample mean is 84.4 and in another sample of 10 observations it was 102.6. Test at 5% level whether the populations have the same variance.

Solution:- Let  $\sigma_1^2$  and  $\sigma_2^2$  be the variances of the two normal populations from which the samples are drawn.

Let the null hypothesis be  $H_0: \sigma_1^2 = \sigma_2^2$

Then the alternative hypothesis be  $H_1: \sigma_1^2 \neq \sigma_2^2$

Level of significance  $\alpha = 0.05$

$n_1 = 8$  and  $n_2 = 10$  also  $\sum (x_i - \bar{x})^2 = 84.4$ ,  $\sum (y_i - \bar{y})^2 = 102.6$

If  $S_1^2$  and  $S_2^2$  be the estimates of  $\sigma_1^2$  and  $\sigma_2^2$  then

$$S_1^2 = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1} = \frac{84.4}{8 - 1} = \frac{84.4}{7} = 12.057$$

$$\text{and } S_2^2 = \frac{\sum (y_i - \bar{y})^2}{n_2 - 1} = \frac{102.6}{10 - 1} = \frac{102.6}{9} = 11.4$$

$$\text{Test statistic: } F = \frac{S_1^2}{S_2^2} = \frac{12.057}{11.4} = 1.057$$

ie Calculated  $F = 1.057$

Tabulated  $F_{0.05}(7, 9)$  degrees of freedom is 3.29

ie  $F_{0.05}(7, 9) = 3.29$

Conclusion:-  $F_{cal} = 1.057$  •  $F_{tab} = 3.29$

$F_{cal} < F_{tab} \therefore$  we Accept  $H_0$

$\therefore$  we conclude that the populations have the same variance.

Problem ⑤:- The time taken by workers in performing a ① Job by method I and Method II is given below.

Method-I : 20 16 26 27 23 22 -

Method-II : 27 33 42 35 32 34 38

Do the data show that the Variances of time distribution from population from which these samples are drawn do not differ Significantly?

Solution:- let the null hypothesis  $H_0: \sigma_1^2 = \sigma_2^2$ ,

$\sigma_1^2$  and  $\sigma_2^2$  are the Variances of the Populations

The alternative hypothesis  $H_1: \sigma_1^2 \neq \sigma_2^2$

Level of Significance  $\alpha = 0.05$  (assume)

Calculation of Sample Variances:

$$\bar{x} = \frac{\sum x_i}{n_1} = \frac{20+16+26+27+23+22}{6} = \frac{134}{6} = 22.3$$

$$\bar{y} = \frac{\sum y_i}{n_2} = \frac{27+33+42+35+32+34+38}{7} = \frac{241}{7} = 34.4$$

$$\text{and } \sum (x_i - \bar{x})^2 = (20-22.3)^2 + (16-22.3)^2 + \dots + (22-22.3)^2 = 81.34$$

$$\sum (y_i - \bar{y})^2 = (27-34.4)^2 + (33-34.4)^2 + \dots + (38-34.4)^2 = 133.72$$

If  $s_1^2$  and  $s_2^2$  be the estimates of  $\sigma_1^2$  and  $\sigma_2^2$ , then

$$s_1^2 = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1} = \frac{81.34}{6-1} = \frac{81.34}{5} = 16.26$$

$$s_2^2 = \frac{\sum (y_i - \bar{y})^2}{n_2 - 1} = \frac{133.72}{7-1} = \frac{133.72}{6} = 22.29$$

Since  $s_2^2 > s_1^2$ , then the test statistic

$$F = \frac{s_2^2}{s_1^2} = \frac{22.29}{16.26} = 1.3699 \cong 1.37$$

$$\text{Calculated } F = 1.37$$

Tabulated F for  $(n_2-1, n_1-1)$  d.f i.e (6,5) d.f at 5% Level is 4.95  
i.e Tabulated  $F = 4.95$

Conclusion:  $F_{cal} = 1.37$  and  $F_{tab} = 4.95$

$$F_{cal} < F_{tab} \therefore \text{We Accept } H_0$$

We Conclude that, There is no Significant difference between the Variances of the time distribution by the workers.

Problem 4:- The nicotine Contents in milligrams in two Samples of tobacco were found to be as follows.

Sample A : 24 27 26 21 25 -

Sample B : 27 30 28 31 22 36

Can it be said that the two Samples have come from the same normal population?

Solution:- To test whether the two Samples came from same normal population, We have to test (1) The equality of Variances by using F-test (2) The equality of means by using Student's t-test.

Given  $n_1 = 5$   $n_2 = 6$

Calculation for means and S.D's of the samples.

$$\bar{x} = \frac{\sum x_i}{n_1} = \frac{24+27+26+21+25}{5} = \frac{123}{5} = 24.6$$

$$\bar{y} = \frac{\sum y_i}{n_2} = \frac{27+30+28+31+22+36}{6} = \frac{174}{6} = 29$$

$$\sum (x_i - \bar{x})^2 = (24-24.6)^2 + (27-24.6)^2 + \dots + (25-24.6)^2 = 21.2$$

$$\sum (y_i - \bar{y})^2 = (27-29)^2 + (30-29)^2 + \dots + (36-29)^2 = 108$$

$$\therefore S_1^2 = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1} = \frac{21.2}{4} = 5.3$$

$$S_2^2 = \frac{\sum (y_i - \bar{y})^2}{n_2 - 1} = \frac{108}{5} = 21.6$$

- (1) F-test:-
- 1) Null hypothesis  $H_0: \sigma_1^2 = \sigma_2^2$  (The Variances are equal)
  - 2) Alternative hypothesis  $H_1: \sigma_1^2 \neq \sigma_2^2$
  - 3) Level of Significance  $\alpha = 0.05$  (assume)
  - 4) Test statistic:  $F = \frac{s_2^2}{s_1^2} = \frac{21.6}{5.3} = 4.075$  ( $s_2^2 > s_1^2$ )

$$\therefore \text{calculated } F = 4.075$$

Tabulated F for  $(n_2-1, n_1-1)$  d.f ie, (5, 4) d.f at 5% level is 6.26

$$\therefore \text{Tabulated } F = 6.26$$

(5) Conclusion:  $F_{cal} = 4.075$ ;  $F_{tab} = 6.26$

$F_{cal} < F_{tab} \therefore$  We Accept  $H_0$

$\therefore$  We conclude that the ~~variances~~ variances are equal.

- (2) Student's t-test:-
- (1) Null hypothesis  $H_0: \mu_1 = \mu_2$  (ie The means are equal)
  - (2) Alternative hypothesis  $H_1: \mu_1 \neq \mu_2$  (two tailed test)
  - (3) Level of Significance  $\alpha = 0.05$  (assume)
- $t_{\alpha}$  value for two tailed test at  $\alpha = 0.05$  is with  $n_1 + n_2 - 2$  ie  $(5+6-2) = 9$  d.f is 2.26 ie  $t_{\alpha} = 2.26$

(4) Test statistic:  $t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

$$s^2 = \frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2} = \frac{21.2 + 108}{5 + 6 - 2} = \frac{129.2}{9}$$

$$s^2 = 14.35$$

$$s = \sqrt{14.35} = 3.79$$

$$t = \frac{24.6 - 29}{(3.79) \sqrt{\frac{1}{5} + \frac{1}{6}}} = -1.92$$

$$|t| = 1.92$$

Conclusion:-  $|t| = 1.92$ ;  $t_{\alpha} = 2.26$

$|t| < t_{\alpha} \therefore$  we Accept  $H_0$

$\therefore$  We conclude that the two means are equal.

$\therefore$  From (i) and (ii) we conclude that the two samples came from same normal populations ( $\mu_1 = \mu_2$  &  $\sigma_1^2 = \sigma_2^2$ )

Practice Problems:- (1) Pumpkins were grown under two experimental conditions. two random samples of 11 and 9 pumpkins, show the sample standard deviations of their weights as 0.8 and 0.5 respectively. Assuming that the weight distributions are normal, test hypothesis that the true variances are equal.

(2) The measurements of the output of two units have given the following results. Assuming that both samples have been obtained from the normal populations at 50% significant level, test whether the two populations have the same variance?

Unit-A : 14.1 10.1 14.7 13.7 14.0

Unit-B : 14.0 14.5 13.7 12.7 14.1

(3) Two random samples gave the following results:

| Sample | Size | Sample mean | Sum of Squares of deviations from the mean |
|--------|------|-------------|--------------------------------------------|
| 1      | 10   | 15          | 90                                         |
| 2      | 12   | 14          | 108                                        |

Test whether the samples came from the same normal population.

(4) Two random samples drawn from normal populations are:

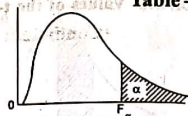
Sample-I : 20 16 26 27 23 22 18 24 25 19 - -

Sample-II : 27 33 42 35 32 34 38 28 41 43 39 37

Obtain the estimates of the variances of populations. Test whether the two populations have the same variance.

Critical Values of the F-Distribution

Table - 5



| Values of $F_{0.05}(v_1, v_2)$ |       |       |       |       |       |       |       |       |       |
|--------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $v_2$                          | $v_1$ |       |       |       |       |       |       |       |       |
|                                | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     |
| 1                              | 161.4 | 199.5 | 215.7 | 224.6 | 230.2 | 234.0 | 236.8 | 238.9 | 240.5 |
| 2                              | 18.51 | 19.00 | 19.16 | 19.25 | 19.30 | 19.33 | 19.35 | 19.37 | 19.38 |
| 3                              | 10.13 | 9.55  | 9.28  | 9.12  | 9.01  | 8.94  | 8.89  | 8.85  | 8.81  |
| 4                              | 7.71  | 6.94  | 6.59  | 6.39  | 6.26  | 6.16  | 6.09  | 6.04  | 6.00  |
| 5                              | 6.61  | 5.79  | 5.41  | 5.19  | 5.05  | 4.95  | 4.88  | 4.82  | 4.77  |
| 6                              | 5.99  | 5.14  | 4.76  | 4.53  | 4.39  | 4.28  | 4.21  | 4.15  | 4.10  |
| 7                              | 5.59  | 4.74  | 4.35  | 4.12  | 3.97  | 3.87  | 3.79  | 3.73  | 3.68  |
| 8                              | 5.32  | 4.46  | 4.07  | 3.84  | 3.69  | 3.58  | 3.50  | 3.44  | 3.39  |
| 9                              | 5.12  | 4.26  | 3.86  | 3.63  | 3.48  | 3.37  | 3.29  | 3.23  | 3.18  |
| 10                             | 4.96  | 4.10  | 3.71  | 3.48  | 3.33  | 3.22  | 3.14  | 3.07  | 3.02  |
| 11                             | 4.84  | 3.98  | 3.59  | 3.36  | 3.20  | 3.09  | 3.01  | 2.95  | 2.90  |
| 12                             | 4.75  | 3.89  | 3.49  | 3.26  | 3.11  | 3.00  | 2.91  | 2.85  | 2.80  |
| 13                             | 4.67  | 3.81  | 3.41  | 3.18  | 3.03  | 2.92  | 2.83  | 2.77  | 2.71  |
| 14                             | 4.60  | 3.74  | 3.34  | 3.11  | 2.96  | 2.85  | 2.76  | 2.70  | 2.65  |
| 15                             | 4.54  | 3.68  | 3.29  | 3.06  | 2.90  | 2.79  | 2.71  | 2.64  | 2.59  |
| 16                             | 4.49  | 3.63  | 3.24  | 3.01  | 2.85  | 2.74  | 2.66  | 2.59  | 2.54  |
| 17                             | 4.45  | 3.59  | 3.20  | 2.96  | 2.81  | 2.70  | 2.61  | 2.55  | 2.49  |
| 18                             | 4.41  | 3.55  | 3.16  | 2.93  | 2.77  | 2.66  | 2.58  | 2.51  | 2.46  |
| 19                             | 4.38  | 3.52  | 3.13  | 2.90  | 2.74  | 2.63  | 2.54  | 2.48  | 2.42  |
| 20                             | 4.35  | 3.49  | 3.10  | 2.87  | 2.71  | 2.60  | 2.51  | 2.45  | 2.39  |
| 21                             | 4.32  | 3.47  | 3.07  | 2.84  | 2.68  | 2.57  | 2.49  | 2.42  | 2.37  |
| 22                             | 4.30  | 3.44  | 3.05  | 2.82  | 2.66  | 2.55  | 2.46  | 2.40  | 2.34  |
| 23                             | 4.28  | 3.42  | 3.03  | 2.80  | 2.64  | 2.53  | 2.44  | 2.37  | 2.32  |
| 24                             | 4.26  | 3.40  | 3.01  | 2.78  | 2.62  | 2.51  | 2.42  | 2.36  | 2.30  |
| 25                             | 4.24  | 3.39  | 2.99  | 2.76  | 2.60  | 2.49  | 2.40  | 2.34  | 2.28  |
| 26                             | 4.23  | 3.37  | 2.98  | 2.74  | 2.59  | 2.47  | 2.39  | 2.32  | 2.27  |
| 27                             | 4.21  | 3.35  | 2.96  | 2.73  | 2.57  | 2.46  | 2.37  | 2.31  | 2.25  |
| 28                             | 4.20  | 3.34  | 2.95  | 2.71  | 2.56  | 2.45  | 2.36  | 2.29  | 2.24  |
| 29                             | 4.18  | 3.33  | 2.93  | 2.70  | 2.55  | 2.43  | 2.35  | 2.28  | 2.22  |
| 30                             | 4.17  | 3.32  | 2.92  | 2.69  | 2.53  | 2.42  | 2.33  | 2.27  | 2.21  |
| 40                             | 4.08  | 3.23  | 2.84  | 2.61  | 2.45  | 2.34  | 2.25  | 2.18  | 2.12  |
| 60                             | 4.00  | 3.15  | 2.76  | 2.53  | 2.37  | 2.25  | 2.17  | 2.10  | 2.04  |
| 120                            | 3.92  | 3.07  | 2.68  | 2.45  | 2.29  | 2.17  | 2.09  | 2.02  | 1.96  |
| ∞                              | 3.84  | 3.00  | 2.60  | 2.37  | 2.21  | 2.10  | 2.01  | 1.94  | 1.88  |

(Continued) Critical Values of the F-Distribution

| Values of $F_{0.05}(v_1, v_2)$ |       |       |       |       |       |       |       |       |       |       |  |  |
|--------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
| $v_2$                          | $v_1$ |       |       |       |       |       |       |       |       |       |  |  |
|                                | 10    | 12    | 15    | 20    | 24    | 30    | 40    | 60    | 120   | ∞     |  |  |
| 1                              | 241.9 | 243.9 | 245.9 | 248.0 | 249.1 | 250.1 | 251.1 | 252.2 | 253.3 | 254.3 |  |  |
| 2                              | 19.40 | 19.41 | 19.43 | 19.45 | 19.45 | 19.46 | 19.47 | 19.48 | 19.49 | 19.50 |  |  |
| 3                              | 8.79  | 8.74  | 8.70  | 8.66  | 8.64  | 8.62  | 8.59  | 8.57  | 8.55  | 8.52  |  |  |
| 4                              | 5.96  | 5.91  | 5.86  | 5.80  | 5.77  | 5.75  | 5.72  | 5.69  | 5.66  | 5.62  |  |  |
| 5                              | 4.74  | 4.68  | 4.62  | 4.56  | 4.53  | 4.50  | 4.46  | 4.43  | 4.40  | 4.36  |  |  |
| 6                              | 4.06  | 4.00  | 3.94  | 3.87  | 3.84  | 3.81  | 3.77  | 3.74  | 3.70  | 3.67  |  |  |
| 7                              | 3.64  | 3.57  | 3.51  | 3.44  | 3.41  | 3.38  | 3.34  | 3.30  | 3.27  | 3.23  |  |  |
| 8                              | 3.35  | 3.28  | 3.22  | 3.15  | 3.12  | 3.08  | 3.04  | 3.01  | 2.97  | 2.93  |  |  |
| 9                              | 3.14  | 3.07  | 3.01  | 2.94  | 2.90  | 2.86  | 2.83  | 2.79  | 2.75  | 2.71  |  |  |
| 10                             | 2.98  | 2.91  | 2.85  | 2.77  | 2.74  | 2.70  | 2.66  | 2.62  | 2.58  | 2.54  |  |  |
| 11                             | 2.85  | 2.79  | 2.72  | 2.65  | 2.61  | 2.57  | 2.53  | 2.49  | 2.45  | 2.40  |  |  |
| 12                             | 2.75  | 2.69  | 2.62  | 2.54  | 2.51  | 2.47  | 2.43  | 2.38  | 2.34  | 2.30  |  |  |
| 13                             | 2.67  | 2.60  | 2.53  | 2.46  | 2.42  | 2.38  | 2.34  | 2.30  | 2.25  | 2.21  |  |  |
| 14                             | 2.60  | 2.53  | 2.46  | 2.39  | 2.35  | 2.31  | 2.27  | 2.22  | 2.18  | 2.13  |  |  |
| 15                             | 2.54  | 2.48  | 2.40  | 2.33  | 2.29  | 2.25  | 2.20  | 2.16  | 2.11  | 2.07  |  |  |
| 16                             | 2.49  | 2.42  | 2.35  | 2.28  | 2.24  | 2.19  | 2.15  | 2.11  | 2.06  | 2.01  |  |  |
| 17                             | 2.45  | 2.38  | 2.31  | 2.23  | 2.19  | 2.17  | 2.10  | 2.06  | 2.01  | 1.96  |  |  |
| 18                             | 2.41  | 2.34  | 2.27  | 2.19  | 2.15  | 2.11  | 2.06  | 2.02  | 1.97  | 1.92  |  |  |
| 19                             | 2.38  | 2.31  | 2.23  | 2.16  | 2.11  | 2.07  | 2.03  | 1.98  | 1.93  | 1.88  |  |  |
| 20                             | 2.35  | 2.28  | 2.20  | 2.12  | 2.08  | 2.04  | 1.99  | 1.95  | 1.90  | 1.84  |  |  |
| 21                             | 2.32  | 2.25  | 2.18  | 2.10  | 2.05  | 2.01  | 1.96  | 1.92  | 1.87  | 1.81  |  |  |
| 22                             | 2.30  | 2.23  | 2.15  | 2.07  | 2.03  | 1.98  | 1.94  | 1.89  | 1.84  | 1.78  |  |  |
| 23                             | 2.27  | 2.20  | 2.13  | 2.05  | 2.01  | 1.96  | 1.91  | 1.86  | 1.81  | 1.76  |  |  |
| 24                             | 2.25  | 2.18  | 2.11  | 2.03  | 1.98  | 1.94  | 1.89  | 1.84  | 1.79  | 1.73  |  |  |
| 25                             | 2.24  | 2.16  | 2.09  | 2.01  | 1.96  | 1.92  | 1.87  | 1.82  | 1.77  | 1.71  |  |  |
| 26                             | 2.22  | 2.15  | 2.07  | 1.99  | 1.95  | 1.90  | 1.85  | 1.80  | 1.75  | 1.69  |  |  |
| 27                             | 2.20  | 2.13  | 2.06  | 1.97  | 1.93  | 1.88  | 1.84  | 1.79  | 1.73  | 1.67  |  |  |
| 28                             | 2.19  | 2.12  | 2.04  | 1.96  | 1.91  | 1.87  | 1.82  | 1.77  | 1.71  | 1.65  |  |  |
| 29                             | 2.18  | 2.10  | 2.03  | 1.94  | 1.90  | 1.85  | 1.81  | 1.75  | 1.70  | 1.64  |  |  |
| 30                             | 2.16  | 2.09  | 2.01  | 1.93  | 1.89  | 1.84  | 1.79  | 1.75  | 1.68  | 1.62  |  |  |
| 40                             | 2.08  | 2.00  | 1.92  | 1.84  | 1.79  | 1.74  | 1.69  | 1.64  | 1.58  | 1.51  |  |  |
| 60                             | 1.99  | 1.92  | 1.84  | 1.75  | 1.70  | 1.65  | 1.59  | 1.53  | 1.47  | 1.39  |  |  |
| 120                            | 1.91  | 1.83  | 1.75  | 1.66  | 1.61  | 1.55  | 1.50  | 1.43  | 1.35  | 1.25  |  |  |
| ∞                              | 1.83  | 1.75  | 1.67  | 1.57  | 1.52  | 1.46  | 1.39  | 1.32  | 1.22  | 1.00  |  |  |

(Continued) Critical Values of the F-Distribution

|          |       | Values of $F_{0.01}(v_1, v_2)$ |       |       |       |       |       |       |       |   |
|----------|-------|--------------------------------|-------|-------|-------|-------|-------|-------|-------|---|
|          |       | $v_1$                          |       |       |       |       |       |       |       |   |
| $v_2$    |       | 1                              | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9 |
| 1        | 4052  | 4999.5                         | 5403  | 5625  | 5764  | 5859  | 5928  | 5981  | 6022  |   |
| 2        | 98.50 | 99.00                          | 99.17 | 99.25 | 99.30 | 99.33 | 99.36 | 99.37 | 99.39 |   |
| 3        | 34.12 | 30.82                          | 29.46 | 28.71 | 28.24 | 27.91 | 27.67 | 27.49 | 27.35 |   |
| 4        | 21.20 | 18.00                          | 16.69 | 15.98 | 15.52 | 15.21 | 14.98 | 14.80 | 14.66 |   |
| 5        | 16.26 | 13.27                          | 12.06 | 11.39 | 10.97 | 10.67 | 10.46 | 10.29 | 10.16 |   |
| 6        | 13.75 | 10.92                          | 9.78  | 9.15  | 8.75  | 8.47  | 8.26  | 8.10  | 7.98  |   |
| 7        | 12.25 | 9.55                           | 8.45  | 7.85  | 7.46  | 7.19  | 6.99  | 6.84  | 6.72  |   |
| 8        | 11.26 | 8.65                           | 7.59  | 7.01  | 6.63  | 6.37  | 6.18  | 6.03  | 5.91  |   |
| 9        | 10.56 | 8.02                           | 6.99  | 6.42  | 6.06  | 5.80  | 5.61  | 5.47  | 5.35  |   |
| 10       | 10.04 | 7.56                           | 6.55  | 5.99  | 5.64  | 5.39  | 5.20  | 5.06  | 4.94  |   |
| 11       | 9.65  | 7.21                           | 6.22  | 5.67  | 5.32  | 5.07  | 4.89  | 4.74  | 4.63  |   |
| 12       | 9.33  | 6.93                           | 5.95  | 5.41  | 5.06  | 4.82  | 4.64  | 4.50  | 4.39  |   |
| 13       | 9.07  | 6.70                           | 5.74  | 5.21  | 4.86  | 4.62  | 4.44  | 4.30  | 4.19  |   |
| 14       | 8.86  | 6.51                           | 5.56  | 5.04  | 4.69  | 4.46  | 4.28  | 4.14  | 4.03  |   |
| 15       | 8.68  | 6.36                           | 5.42  | 4.89  | 4.56  | 4.32  | 4.14  | 4.00  | 3.89  |   |
| 16       | 8.53  | 6.23                           | 5.29  | 4.77  | 4.44  | 4.20  | 4.03  | 3.89  | 3.78  |   |
| 17       | 8.40  | 6.11                           | 5.18  | 4.67  | 4.34  | 4.10  | 3.93  | 3.79  | 3.68  |   |
| 18       | 8.29  | 6.01                           | 5.09  | 4.58  | 4.25  | 4.01  | 3.84  | 3.71  | 3.60  |   |
| 19       | 8.18  | 5.93                           | 5.01  | 4.50  | 4.17  | 3.94  | 3.77  | 3.63  | 3.52  |   |
| 20       | 8.10  | 5.85                           | 4.94  | 4.43  | 4.10  | 3.87  | 3.70  | 3.56  | 3.46  |   |
| 21       | 8.02  | 5.78                           | 4.87  | 4.37  | 4.04  | 3.81  | 3.64  | 3.51  | 3.40  |   |
| 22       | 7.95  | 5.72                           | 4.82  | 4.31  | 3.99  | 3.76  | 3.59  | 3.45  | 3.35  |   |
| 23       | 7.88  | 5.66                           | 4.76  | 4.26  | 3.94  | 3.71  | 3.54  | 3.41  | 3.30  |   |
| 24       | 7.82  | 5.61                           | 4.72  | 4.22  | 3.90  | 3.67  | 3.50  | 3.36  | 3.26  |   |
| 25       | 7.77  | 5.57                           | 4.68  | 4.18  | 3.85  | 3.63  | 3.46  | 3.32  | 3.22  |   |
| 26       | 7.72  | 5.53                           | 4.64  | 4.14  | 3.82  | 3.59  | 3.42  | 3.29  | 3.18  |   |
| 27       | 7.68  | 5.49                           | 4.60  | 4.11  | 3.78  | 3.56  | 3.39  | 3.26  | 3.15  |   |
| 28       | 7.64  | 5.45                           | 4.57  | 4.07  | 3.75  | 3.53  | 3.36  | 3.23  | 3.12  |   |
| 29       | 7.60  | 5.42                           | 4.54  | 4.04  | 3.73  | 3.50  | 3.33  | 3.20  | 3.09  |   |
| 30       | 7.56  | 5.39                           | 4.51  | 4.02  | 3.70  | 3.47  | 3.30  | 3.17  | 3.07  |   |
| 40       | 7.31  | 5.18                           | 4.31  | 3.83  | 3.51  | 3.29  | 3.12  | 2.99  | 2.89  |   |
| 60       | 7.08  | 4.98                           | 4.13  | 3.65  | 3.34  | 3.12  | 2.95  | 2.82  | 2.72  |   |
| 120      | 6.85  | 4.79                           | 3.95  | 3.48  | 3.17  | 2.96  | 2.79  | 2.66  | 2.56  |   |
| $\infty$ | 6.63  | 4.61                           | 3.78  | 3.32  | 3.02  | 2.80  | 2.64  | 2.51  | 2.41  |   |

(Continued) Critical Values of the F-Distribution

|          |       | Values of $F_{0.01}(v_1, v_2)$ |       |       |       |       |       |       |       |       |          |  |
|----------|-------|--------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|----------|--|
|          |       | $v_1$                          |       |       |       |       |       |       |       |       |          |  |
| $v_2$    |       | 10                             | 12    | 15    | 20    | 24    | 30    | 40    | 60    | 120   | $\infty$ |  |
| 1        | 6056  | 6106                           | 6157  | 6209  | 6235  | 6261  | 6287  | 6313  | 6339  | 6366  |          |  |
| 2        | 99.40 | 99.42                          | 99.43 | 99.45 | 99.46 | 99.47 | 99.47 | 99.48 | 99.49 | 99.50 |          |  |
| 3        | 27.23 | 27.05                          | 26.87 | 26.69 | 26.60 | 26.50 | 26.41 | 26.32 | 26.22 | 26.13 |          |  |
| 4        | 14.55 | 14.37                          | 14.20 | 14.02 | 13.93 | 13.84 | 13.75 | 13.65 | 13.56 | 13.46 |          |  |
| 5        | 10.05 | 9.89                           | 9.72  | 9.55  | 9.47  | 9.38  | 9.29  | 9.20  | 9.11  | 9.02  |          |  |
| 6        | 7.87  | 7.72                           | 7.56  | 7.40  | 7.31  | 7.23  | 7.14  | 7.06  | 6.97  | 6.88  |          |  |
| 7        | 6.62  | 6.47                           | 6.31  | 6.16  | 6.07  | 5.99  | 5.91  | 5.82  | 5.74  | 5.65  |          |  |
| 8        | 5.81  | 5.67                           | 5.52  | 5.36  | 5.28  | 5.20  | 5.12  | 5.03  | 4.95  | 4.86  |          |  |
| 9        | 5.26  | 5.11                           | 4.96  | 4.81  | 4.73  | 4.65  | 4.57  | 4.48  | 4.40  | 4.31  |          |  |
| 10       | 4.85  | 4.71                           | 4.56  | 4.41  | 4.33  | 4.25  | 4.17  | 4.08  | 4.00  | 3.91  |          |  |
| 11       | 4.54  | 4.40                           | 4.25  | 4.10  | 4.02  | 3.94  | 3.86  | 3.78  | 3.69  | 3.60  |          |  |
| 12       | 4.30  | 4.16                           | 4.01  | 3.86  | 3.78  | 3.70  | 3.62  | 3.54  | 3.45  | 3.36  |          |  |
| 13       | 4.10  | 3.96                           | 3.82  | 3.66  | 3.59  | 3.51  | 3.43  | 3.34  | 3.25  | 3.17  |          |  |
| 14       | 3.94  | 3.80                           | 3.66  | 3.51  | 3.43  | 3.35  | 3.27  | 3.18  | 3.09  | 3.00  |          |  |
| 15       | 3.80  | 3.67                           | 3.52  | 3.37  | 3.29  | 3.21  | 3.13  | 3.05  | 2.96  | 2.87  |          |  |
| 16       | 3.69  | 3.55                           | 3.41  | 3.26  | 3.18  | 3.10  | 3.02  | 2.93  | 2.84  | 2.75  |          |  |
| 17       | 3.59  | 3.46                           | 3.31  | 3.16  | 3.08  | 3.00  | 2.92  | 2.84  | 2.75  | 2.66  |          |  |
| 18       | 3.51  | 3.37                           | 3.23  | 3.08  | 2.99  | 2.92  | 2.84  | 2.76  | 2.67  | 2.58  |          |  |
| 19       | 3.43  | 3.30                           | 3.15  | 3.00  | 2.92  | 2.84  | 2.76  | 2.67  | 2.58  | 2.49  |          |  |
| 20       | 3.37  | 3.23                           | 3.09  | 2.94  | 2.86  | 2.78  | 2.69  | 2.61  | 2.52  | 2.42  |          |  |
| 21       | 3.31  | 3.17                           | 3.03  | 2.88  | 2.80  | 2.72  | 2.64  | 2.55  | 2.46  | 2.36  |          |  |
| 22       | 3.26  | 3.12                           | 2.98  | 2.83  | 2.75  | 2.67  | 2.58  | 2.50  | 2.40  | 2.31  |          |  |
| 23       | 3.21  | 3.07                           | 2.93  | 2.78  | 2.70  | 2.62  | 2.54  | 2.45  | 2.35  | 2.26  |          |  |
| 24       | 3.17  | 3.03                           | 2.89  | 2.74  | 2.66  | 2.58  | 2.49  | 2.40  | 2.31  | 2.21  |          |  |
| 25       | 3.13  | 2.99                           | 2.85  | 2.70  | 2.62  | 2.54  | 2.45  | 2.36  | 2.27  | 2.17  |          |  |
| 26       | 3.09  | 2.96                           | 2.81  | 2.66  | 2.58  | 2.50  | 2.42  | 2.33  | 2.23  | 2.13  |          |  |
| 27       | 3.06  | 2.93                           | 2.78  | 2.63  | 2.55  | 2.47  | 2.38  | 2.29  | 2.20  | 2.10  |          |  |
| 28       | 3.03  | 2.90                           | 2.75  | 2.60  | 2.52  | 2.44  | 2.35  | 2.26  | 2.17  | 2.06  |          |  |
| 29       | 3.00  | 2.87                           | 2.73  | 2.57  | 2.49  | 2.41  | 2.33  | 2.23  | 2.14  | 2.03  |          |  |
| 30       | 2.98  | 2.84                           | 2.70  | 2.55  | 2.47  | 2.39  | 2.30  | 2.21  | 2.11  | 2.01  |          |  |
| 40       | 2.80  | 2.66                           | 2.52  | 2.37  | 2.29  | 2.20  | 2.11  | 2.02  | 1.92  | 1.80  |          |  |
| 60       | 2.63  | 2.50                           | 2.35  | 2.20  | 2.12  | 2.03  | 1.94  | 1.84  | 1.73  | 1.60  |          |  |
| 120      | 2.47  | 2.34                           | 2.19  | 2.03  | 1.95  | 1.86  | 1.76  | 1.66  | 1.53  | 1.38  |          |  |
| $\infty$ | 2.32  | 2.18                           | 2.04  | 1.88  | 1.79  | 1.70  | 1.59  | 1.47  | 1.32  | 1.00  |          |  |